

Photon Transfer Curve — ThorCam Noise Decomposition

Setup

Following the two-frame-difference method from Janesick's *Photon Transfer* (SPIE), we recorded bright/dark frame stacks across 20 illumination levels spanning 0.1%–99.9% of full well, at fixed gain (index 0), with the ThorCam (serial 35596). For each level: 50 bright frames (filter wheel open) and 50 dark frames (filter wheel closed, back-to-back), both cropped to the central 70×70 px sub-ROI (verified vignetting-free). Exposure at each level was set by autoexposure. Acquisition:

[scripts/test/thorcam_noise_bright.py](#).

All figures and numbers below are reproduced from

[thorcam_noise_analysis.ipynb](#).

Parameter	Value
Gain index	0
Signal levels	20 (0.1%–99.9% of full well)
Frames per level	50 (bright), 50 (dark)
ROI	central 70×70 px
ADC full-scale (<code>pixel_max</code>)	4095 DN
Dark subtraction	master dark (mean of 50 dark frames per level)

Noise Model

Total pixel noise decomposes into three independent terms (Janesick, Ch. 4):

$$\text{sigma_total}^2 = \text{sigma_read}^2 + \text{sigma_shot}^2 + \text{sigma_fpn}^2$$

- **sigma_read** — read noise, DN, signal-independent (flat floor).
- **sigma_shot** = $\sqrt{S / K}$ — photon shot noise, DN, where **S** is mean signal (DN) and **K** is the conversion gain (e⁻/DN). Slope-½ on a log-log plot.
- **sigma_fpn** = **P_N** * **S** — fixed-pattern noise, proportional to signal (**P_N** = PRNU factor). Slope-1, dominates near full well.

`sigma_shot` and `sigma_fpn` are both signal-dependent but statistically independent (one temporal/random, one fixed/spatial), so they don't separate in a single frame's spatial std. The standard trick is the **two-frame difference method**: for a pair of frames `F1`, `F2` at identical illumination,

```
D = F1 - F2
sigma_temporal^2 = var(D) / 2    (= sigma_read^2 + sigma_shot^2)
```

Differencing cancels any *fixed* pattern (it's identical in both frames), and mean-subtracting `D` before taking its variance additionally cancels any *common-mode* fluctuation shared by every pixel — including global lamp flicker between the two frames. `sigma_fpn` is then recovered by quadrature subtraction from the single-frame spatial std, `sigma_total`:

```
sigma_fpn = sqrt(max(sigma_total^2 - sigma_temporal^2, 0))
```

Conversion Gain and Read Noise

`K` is fit from $\text{sigma_temporal}^2 = \text{sigma_read_var} + S/K$, restricted to the **linear shot region** (1%–60% of full well, `S` from ~40 to ~2460 DN). Above ~60% of full well, `var/S` rolls off as the sensor approaches saturation — including that tail biases the fit, so it is excluded (visible in the raw ratio: flat at $\approx 0.376 \text{ DN}^2/\text{DN}$ from 40–2460 DN, then falling toward 0 above 4000 DN as the pixel saturates).

Read noise is measured **directly** from the dark-frame stacks (same difference method, applied to `dark_dn`), not extrapolated from the PTC intercept — the intercept from a linear fit through a signal range that starts at ~40 DN is a long, noisy extrapolation back to zero and is reported only as a diagnostic.

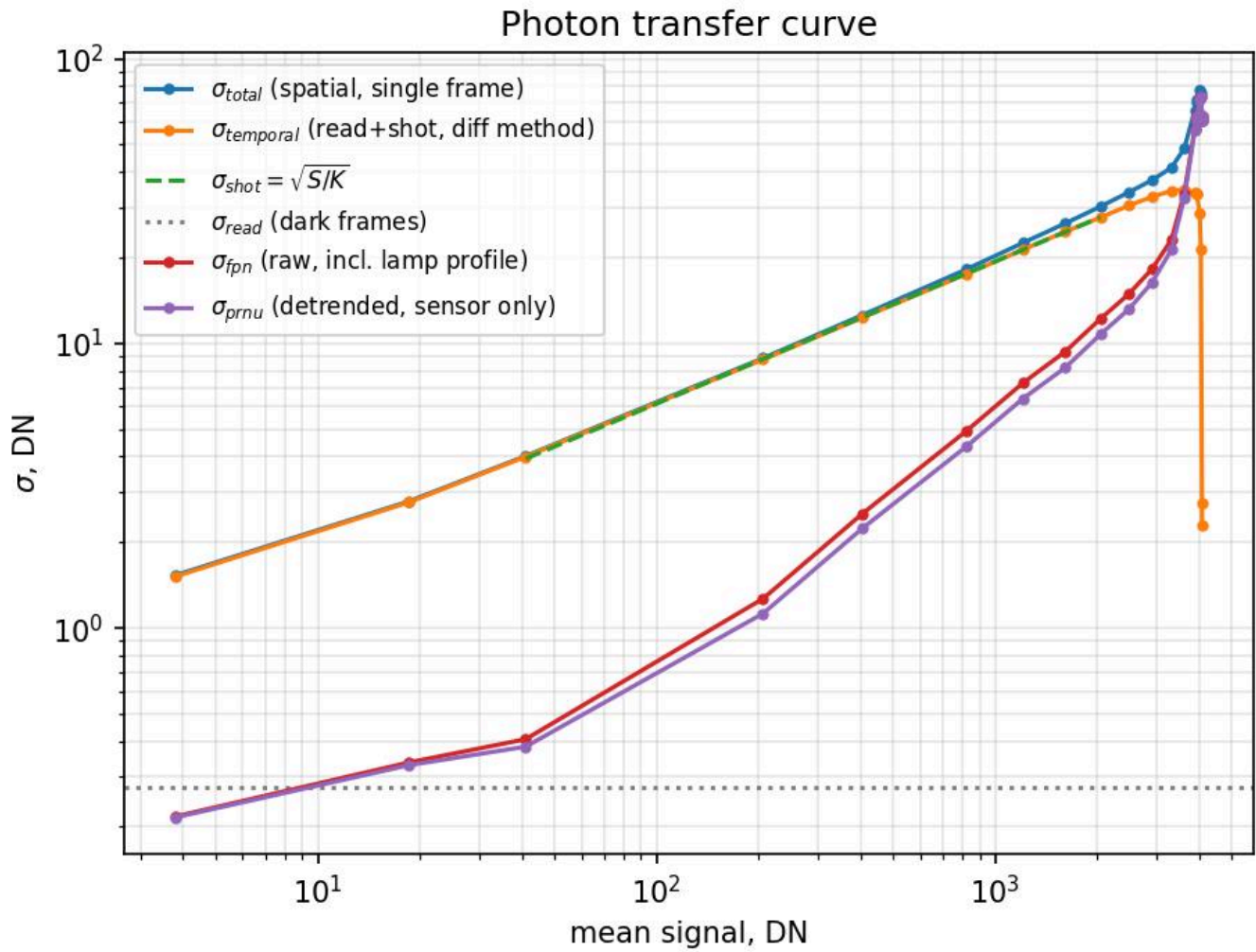


Figure 1. Full noise decomposition: σ_{total} (spatial), σ_{temporal} (difference method), σ_{shot} (fit), σ_{read} (dark-frame floor), σ_{fpn} (raw, quadrature), σ_{prnu} (detrended, quadrature). Log-log.

Quantity	Value	Source
Conversion gain, K	2.660 e ⁻ /DN	this fit (linear region)
Conversion gain, K (cross-check)	2.790 e ⁻ /DN	thorcam_ptc_gain.py , dedicated PTC sweep
Read noise (direct)	0.275 DN (0.73 e ⁻)	dark-frame difference, this dataset
Read noise (cross-check)	0.256 DN	thorcam_read_noise.py , dedicated sweep
Read noise (PTC intercept, diagnostic)	≈0.00 DN	quadrature fit intercept, not used
Full well (pixel_max * K)	≈10,900 e ⁻	ADC-clip upper bound
Dynamic range (pixel_max / read noise)	≈16,000:1 (~84 dB)	quantization-limited

K agrees with the dedicated PTC measurement to ~5%; read noise agrees with the dedicated read-noise sweep to ~7% — both within expected run-to-run variation for this ROI/lighting setup, confirming the fit is well-conditioned once restricted to the linear region.

Fixed-Pattern Noise: Raw vs. Detrended

The raw `sigma_fpn` includes not just sensor PRNU but also the lamp's own spatial illumination profile across the 70×70 ROI (a smooth gradient, not a sensor property). To isolate true pixel-to-pixel PRNU, a low-order 2D polynomial (the smooth illumination shape, fit per level from the mean image) is subtracted before recomputing the spatial std, giving `sigma_prnu`.

Quantity	Value
PRNU factor, <code>P_N</code> (detrended, sensor-only)	≈0.53% of signal
Raw FPN factor (includes lamp profile)	≈0.60% of signal

Detrending only shaves off a modest fraction of the raw FPN — the lamp's spatial non-uniformity across this ROI is small relative to the sensor's own PRNU, so the raw curve is already a reasonable stand-in for PRNU except very close to full well, where the lamp/vignetting gradient becomes more visible (Figure 1, upper right).

Lamp Stability Check

To check whether the two-frame difference method needs to worry about *temporal* lamp flicker (as opposed to the spatial non-uniformity above), the frame-to-frame fluctuation of the ROI mean is compared to what shot+read noise alone would predict.

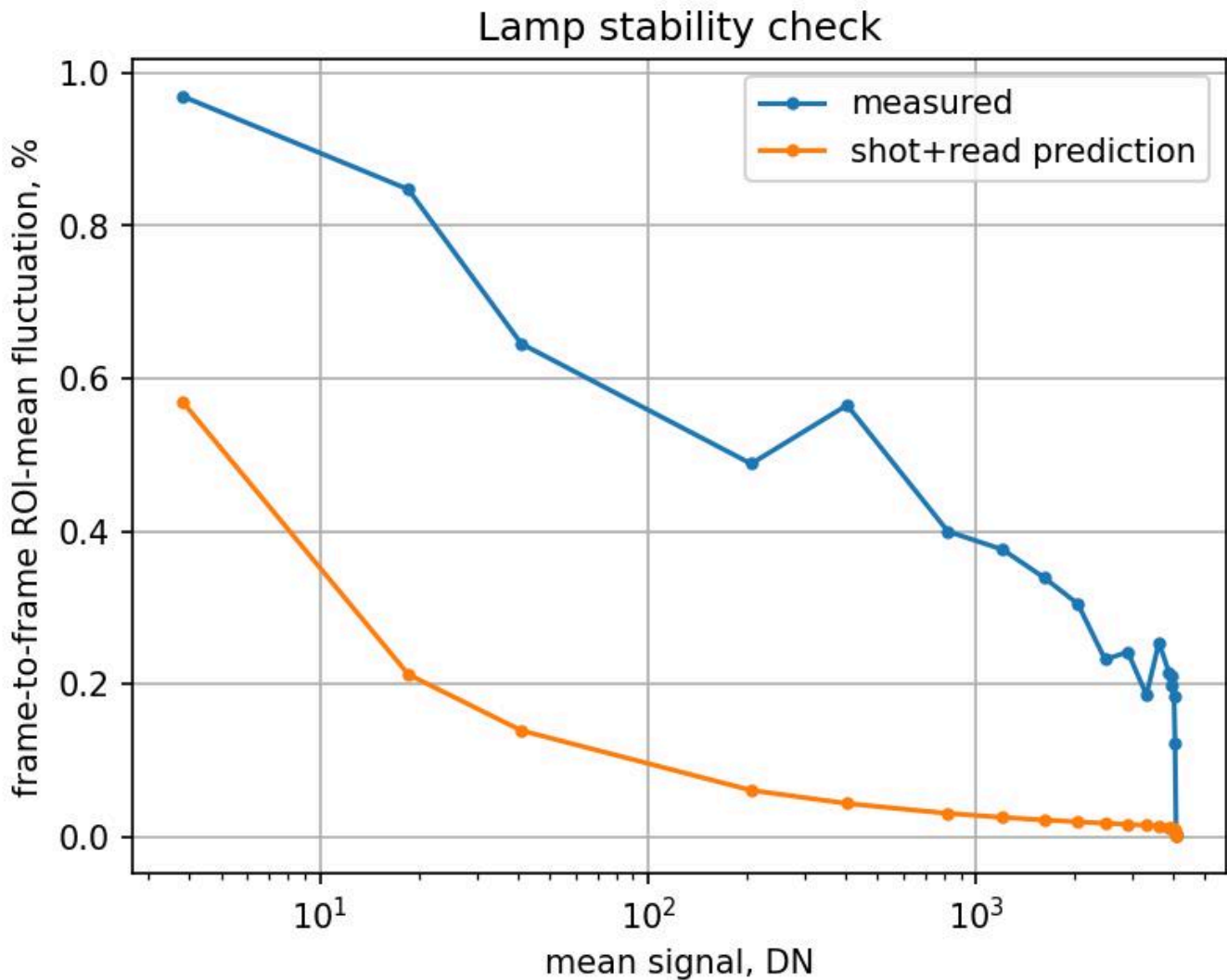


Figure 2. Measured frame-to-frame ROI-mean fluctuation (%) vs. the shot+read-noise prediction, across signal levels.

The measured curve sits above the shot-noise prediction at every level, by a roughly constant **~0.4% excess** at high signal (and more at very low signal, where the percentage is dominated by the read floor). This is the lamp's temporal flicker. It does not corrupt `sigma_temporal` in Figure 1, because mean-subtracting each difference image (`D -= D.mean()`) removes exactly this common-mode fluctuation before the variance is taken — the same technique already used in `thorcam_ptc_gain.py`. It would, however, inflate a naive single-frame-based shot-noise estimate that didn't use the difference method.

Conclusions

- **Lamp noise is real but secondary:** ~0.4% temporal flicker (removed by mean-subtracting each difference pair) and a small spatial illumination gradient across the ROI (removed by polynomial detrending before computing PRNU).

Derived constants (gain index 0, this camera, serial 35596):

Constant	Value
Conversion gain, K	2.66 e⁻/DN (2.79 e ⁻ /DN dedicated PTC)
Read noise, sigma_read	0.275 DN ≈ 0.73 e⁻ (0.256 DN dedicated sweep)
Full well	≈10,900 e⁻ (pixel_max × K , ADC-clip bound)
Dynamic range	≈16,000:1 (~84 dB)
PRNU factor, P_N (sensor only)	≈0.53% of signal
Raw FPN factor (incl. lamp profile)	≈0.60% of signal
Lamp temporal instability (excess)	≈0.2–0.4% frame-to-frame at high signal

Fast Error Estimation

Practical estimators with the constants above already plugged in, for quickly checking "is this measurement good enough" without re-deriving anything. S is mean signal in DN, N is the number of frames averaged.

Per-pixel intensity, single ROI

$$\text{eps_int}(S, N) \text{ [\%]} = 100 * \text{sqrt}((0.076 + 0.376*S)/N + (0.0053*S)^2) / S$$

- $0.076 = \text{sigma_read}^2$, $0.376 = 1/K$ (DN²/DN, shot term), $0.0053 = P_N$ (PRNU).
- Shot-limited quick form (valid above ~20 DN, single frame): $\text{eps_int} \approx 61/\text{sqrt}(S) \text{ \%}$.
- **Averaging floor:** the $0.0053*S$ (PRNU) term does not shrink with N — it's a fixed per-pixel gain error, not noise. Beyond a few hundred frames, per-pixel intensity error stalls at **≈0.53%**, no matter how long you average. Only flat-fielding / referencing removes it (next section).

Transmission ratio (the metasurface case)

For $T = S_{\text{sample}} / S_{\text{ref}}$, the fixed per-pixel PRNU term is the *same* multiplicative gain in both S_{sample} and S_{ref} and **cancels in the ratio** — there is no 0.53% floor here, averaging always keeps helping. The error is purely temporal (shot+read):

$$\text{eps_T}(S_s, N_s, S_r, N_r) \text{ [\%]} = 100 * \text{sqrt}(1/(K*S_s*N_s) + 1/(K*S_r*N_r))$$

Dim/shot-limited single-term approximation (when the reference is much brighter/better averaged than the sample, so the sample term dominates):

$$\begin{aligned} \text{eps_T [\%]} &\approx 61.3 / \sqrt{S_s * N_s} & (S_s = \text{sample/transmitted signal, DN}) \\ N_s \text{ (for target eps\%)} &\approx 3760 / (S_s * \text{eps}^2) \end{aligned}$$

Quick-reference table

S (DN)	eps_int, 1 frame (%)	eps_T, 1 frame (%)	N for eps_T = 0.5%	N for eps_T = 0.1%
100	6.2	6.1	150	3760
250	3.9	3.9	60	1500
500	2.8	2.7	30	750
1000	2.0	1.9	15	380
2000	1.5	1.4	8	190
3600	1.2	1.0	4	104

Worked example: 90%/5% transmission metasurface

Reference (no sample) exposed near full well, $S_{ref} \approx 3600$ DN. Sample regions at 90% and 5% transmission then sit at $S_s \approx 3240$ DN and $S_s \approx 180$ DN respectively. Using the full two-term formula with $N_s = N_r = N$:

Region	S_s (DN)	N for 0.5%	N for 0.1%
90% transmission	3240	~9	~220
5% transmission	180	~88	~2190

The dimmest region sets the averaging budget. If both regions must hit the same target accuracy in the same acquisition, average by the 5%-transmission requirement (~90 frames for 0.5%, ~2200 frames for 0.1%) — the bright region is already far better than needed at that frame count. If that many frames isn't practical, either accept a looser error bound for the dim regions specifically, or increase exposure time (raises S_s directly, same formula).

Reference recap

$$\begin{aligned} \sigma_{total}^2 &= \sigma_{read}^2 + \sigma_{shot}^2 + \sigma_{fpn}^2 \\ \sigma_{shot} &= \sqrt{S / K} & [DN] \\ \sigma_{fpn} &= P_N * S & [DN] \quad \text{-- fixed, does} \end{aligned}$$

not average down

$$S_{\text{electrons}} = S_{\text{DN}} * K$$